

MCV 4U - Chapter 1: Introduction to Calculus Test

Name _____

Date _____

1. Evaluate the following limits.

a) $\lim_{x \rightarrow 0} \frac{\sqrt{x+3} - \sqrt{3-x}}{x}$

$= \lim_{x \rightarrow 0} \frac{(\sqrt{x+3} - \sqrt{3-x})(\sqrt{x+3} + \sqrt{3-x})}{x(\sqrt{x+3} + \sqrt{3-x})}$

$= \lim_{x \rightarrow 0} \frac{2x}{x(\sqrt{x+3} + \sqrt{3-x})}$

$= \frac{2}{2\sqrt{3}}$

$= \frac{1}{\sqrt{3}}$

b) $\lim_{x \rightarrow 5} \sqrt{5x} - 5$

$= \sqrt{5(5)} - 5$

$= 0$

c) $\lim_{x \rightarrow 0} \frac{(x-27)^{\frac{1}{3}} + 3}{x}$

Let $(x-27)^{\frac{1}{3}} = u$
 $x = u^3 + 27$

as $x \rightarrow 0, u \rightarrow -3$

$= \lim_{u \rightarrow -3} \frac{u+3}{u^3+27}$

$= \lim_{u \rightarrow -3} \frac{u+3}{(u+3)(u^2-3u+9)}$

$= \frac{1}{(-3)^2 - 3(-3) + 9}$

$= \frac{1}{9+9+9}$
 $= \frac{1}{27}$

3/3

3/3

1/1

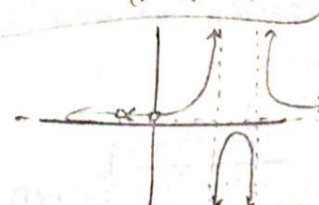
$(a+b)^3 = (a+b)(a^2+ab+b^2)$

2. State the equation of the location of ALL infinite discontinuities in

$f(x) = \frac{x+3}{(x-2)(x^2-9)}$. What would the limit be as x approaches these values?

$S(x) = \frac{x+3}{(x-2)(x-3)(x+3)}$

$S(x) = \frac{1}{(x-2)(x-3)}$



There exist infinite discontinuities/asymptotes at $x=2$ and $x=3$. The limit as x approaches either of these values would not exist.

$\lim_{x \rightarrow 2^-} S(x) = -\infty \neq \lim_{x \rightarrow 2^+} S(x) = \infty \therefore \lim_{x \rightarrow 2} S(x) = \text{DNE}$

$\lim_{x \rightarrow 3^-} S(x) = -\infty \neq \lim_{x \rightarrow 3^+} S(x) = \infty \therefore \lim_{x \rightarrow 3} S(x) = \text{DNE}$

2/2

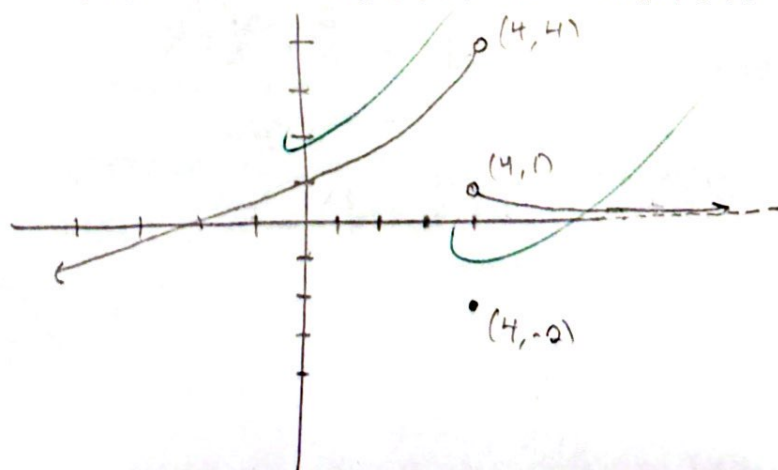
3. Sketch the graph of a function that satisfies the following conditions.

$f(4) = -2$

$\lim_{x \rightarrow 4^-} f(x) = 4$

$\lim_{x \rightarrow 4^+} f(x) = 1$

$\lim_{x \rightarrow \infty} f(x) = 0$



2/2

1/1

4. Fill in the blank to make the function continuous.

A) 3

B) -13

C) 13

D) -3

$$\begin{aligned} &= \frac{(x-1)(x+3)}{-(x+5)} \\ &= -13 \end{aligned}$$

$$f(x) = \begin{cases} x^2 - 7x - 30 & x \neq 10 \\ \frac{10-x}{-13} & x = 10 \end{cases} \quad /1$$

5. Given $\lim_{x \rightarrow 5} f(x) = 2$ and $\lim_{x \rightarrow 5} g(x) = -5$ then

A) $\lim_{x \rightarrow 25} (g(x) \cdot f(x)) = -10$

B) $\lim_{x \rightarrow 10} (g(x) \cdot f(x)) = -10$

C) $\lim_{x \rightarrow 5} (g(x) \cdot f(x)) = -10$

$$\begin{aligned} &\lim_{x \rightarrow 5} g(x) \cdot \lim_{x \rightarrow 5} f(x) \\ &= -5 \cdot 2 = -10 \end{aligned} \quad /1$$

6. Why is $\lim_{x \rightarrow 0} 4\sqrt{x} = 0$ an incorrect statement? Be specific.

The function $4\sqrt{x}$ does not exist when $x < 0$. This is because you cannot take the root of a negative number. In order for $\lim_{x \rightarrow 0} S(x)$ to exist, then $\lim_{x \rightarrow 0} S(x) = \lim_{x \rightarrow 0^+} S(x)$. But since $\lim_{x \rightarrow 0} S(x) = \text{DNE}$, $\lim_{x \rightarrow 0} S(x) \neq \lim_{x \rightarrow 0^+} S(x)$ and $\therefore \lim_{x \rightarrow 0} S(x) = \text{DNE}$. 2 /2

7. Determine the coordinates of a point where the tangent to $f(x) = 3x^2 - 5x - 4$ is perpendicular to the line $x + 3y - 12 = 0$.

$$3y = -x + 12$$

$$y = -\frac{1}{3}x + 4$$

$$m_{\text{line}} = -\frac{1}{3}$$

$$m_{\text{tangent}} = 3$$

$$S(x) = 3x^2 - 5x - 4$$

$$m = \lim_{h \rightarrow 0} \frac{S(x+h) - S(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{[3(x+h)^2 - 5(x+h) - 4] - [3x^2 - 5x - 4]}{h}$$

$$= \lim_{h \rightarrow 0} \frac{3x^2 + 6xh + 3h^2 - 5x - 5h - 4 - (3x^2 - 5x - 4)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h(6x + 3h - 5)}{h}$$

$$= \lim_{h \rightarrow 0} 6x + 3h - 5$$

$$m = 6x - 5$$

$$3 = 6x - 5$$

$$8 = 6x$$

$$x = \frac{4}{3}$$

$$S\left(\frac{4}{3}\right) = 3\left(\frac{4}{3}\right)^2 - 5\left(\frac{4}{3}\right) - 4$$

$$= \frac{48}{9} - \frac{20}{3} - 4$$

$$= \frac{16}{3} - \frac{20}{3} - \frac{12}{3}$$

$$= -\frac{16}{3}$$

$$P\left(\frac{4}{3}, -\frac{16}{3}\right)$$

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8. Suppose you can learn $N(t) = 64t - t^3$ words of a new language after t hours of study time:

- How many words have been learned after 2 hours?
- What is the rate in words per hour at 2 hours?

$$a) N(2) = 64(2) - (2)^3$$

$$= 128 - 8$$

≈ 120 new words have been learned in 2 hours.

b)

$$m = \lim_{h \rightarrow 0} \frac{N(t+h) - N(t)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{[64(t+h) - (t+h)^3] - [64t - t^3]}{h}$$

$$= \lim_{h \rightarrow 0} \frac{[64t + 64h - (t^3 + 3t^2h + 3th^2 + h^3)] - [64t - t^3]}{h}$$

$$= \lim_{h \rightarrow 0} \frac{64h - 3t^2h - 3th^2 - h^3}{h}$$

$$= \lim_{h \rightarrow 0} 64 - 3t^2 - 3th - h^2$$

$$m = 64 - 3t^2$$

$$m = 64 - 3(2)^2$$

$$m = 64 - 12$$

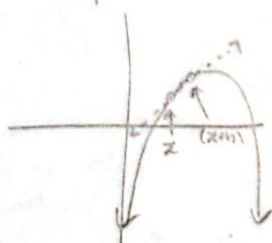
$$m = 52$$

Therefore at 2 hours you are learning at a rate of 52 words/hour.

9. Explain, in detail what is happening graphically when the $\lim_{h \rightarrow 0}$ is added to the formula

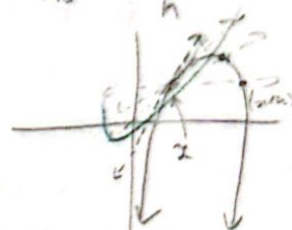
$$g(x) = \frac{f(x+h) - f(x)}{h}$$

$g(x)$ represents the average rate of change of $f(x)$, or the secant.



By adding $\lim_{h \rightarrow 0}$ to the formula: $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

This equation now represents the instantaneous rate of change of $f(x)$, because as $h \rightarrow 0$, the secant approaches the tangent.



10. Determine values of a and b so that the following function is continuous.

For $f(x)$ to be continuous, $\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^+} f(x)$ and $\lim_{x \rightarrow 2} f(x) = f(2)$

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^+} f(x)$$

$$f(x) = \begin{cases} 2 + 3x - 2x^2 & \text{if } x > 2 \\ 2 - x & \text{if } 1 \leq x \leq 2 \\ -3x + 1 & \text{if } x < 1 \end{cases}$$

$$1. \lim_{x \rightarrow 2^-} f(x) \quad 2. \lim_{x \rightarrow 1^-} f(x)$$

$$\lim_{x \rightarrow 2^-} \left[\frac{2 + 3x - 2x^2}{2 - x} \right] \quad \lim_{x \rightarrow 1^-} [-3x + 1]$$

$$= \lim_{x \rightarrow 2^-} \frac{-(2x^2 - 3x - 2)}{2 - x} = -3(1) + 1$$

$$= \lim_{x \rightarrow 2^-} \frac{(2x+1)(x-2)}{2-x} = -2$$

$$= 2(2) + 1 = 5$$

$$P(2, 5)$$

$$P(1, -2)$$

$$f(x) = ax^2 + bx - 2$$

$$5 = a(2)^2 + b(2) - 2$$

$$2b = 7 - 4a$$

$$b = \frac{7 - 4a}{2}$$

$$-2 = a(1)^2 + \left(\frac{7 - 4a}{2} \right)(1) - 2$$

$$a = -\left(\frac{7 - 4a}{2} \right)$$

$$-2a = 7 - 4a$$

$$2a = 7$$

$$a = \frac{7}{2}$$

$$b = \frac{7 - 4\left(\frac{7}{2}\right)}{2}$$

$$b = \frac{7 - 14}{2}$$

$$b = -\frac{7}{2}$$

check

$$\frac{7}{2}(2)^2 - \frac{7}{2}(2) - 2 = 5 \quad \frac{7}{2}(1)^2 - \frac{7}{2}(1) - 2 = -2$$